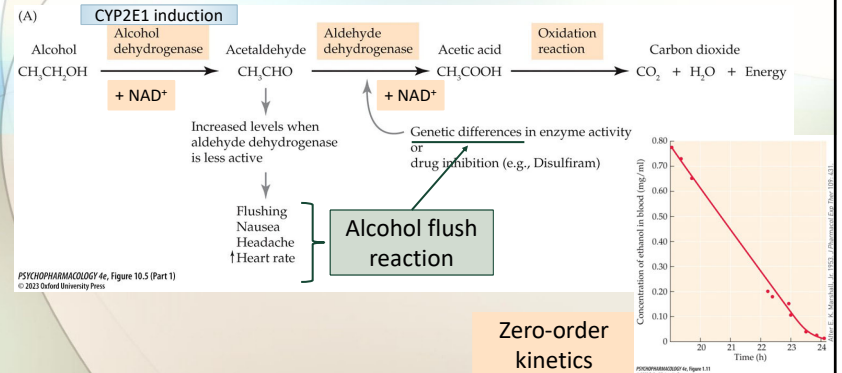


Chapter 10 – Alcohol

Alcohol

Chapter 10

The pharmacokinetics of ethanol: Metabolism



Chronic EtOH use leads to tolerance

Acute

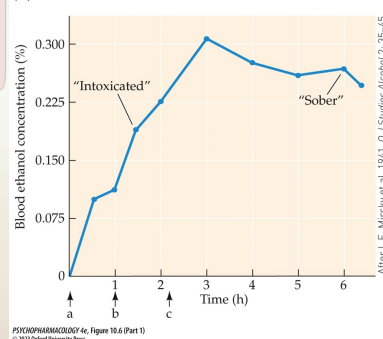
effects of EtOH are reduced during elimination phase

Metabolic

Pharmacodynamic

Behavioral

(A) Acute tolerance



Chronic EtOH use leads to tolerance

Acute

effects of EtOH are reduced during elimination phase

Metabolic

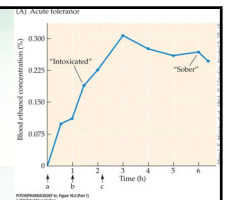
CYP2E1 induction

Pharmacodynamic

changes to Glu, GABA, DA, Opioid receptors

Behavioral

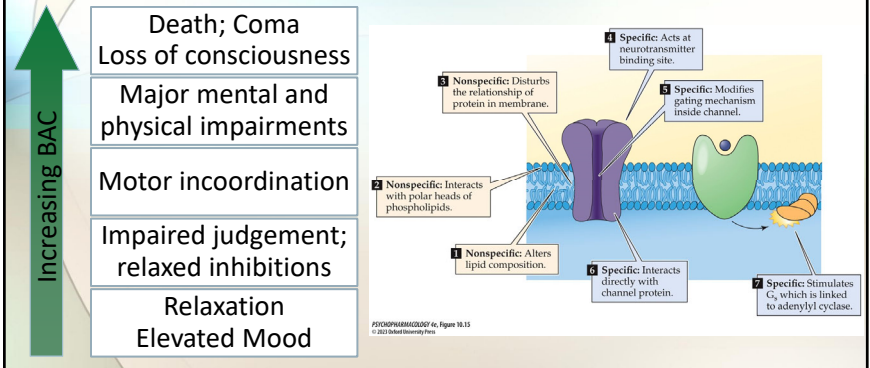
learned adjustments in behaviors practiced under the influence of EtOH



3 Key points

- 1 The drug actions of ethanol
- 2 The behavioral effects of ethanol
- 3 Neuroadaptations following chronic use
 - Glu & GABA receptors
 - Dopamine & Opioid receptors
 - Hypofunction of PFC

EtOH has both specific and non-specific actions



Acute EtOH decreases neuronal excitability

GABA

Glutamate

Acute EtOH decreases neuronal excitability

GABA

- EtOH increases Cl^- flux through GABA_A receptor
- Subunit specificity
 - $\downarrow$ LORR in $\alpha 1/\beta 2$ (synaptic) KO mice
 - $\downarrow$ EtOH consumption in $\alpha 4/\delta$ or $\alpha 6/\delta$ (extrasynaptic) KO mice

Glutamate

Acute EtOH decreases neuronal excitability

Glutamate

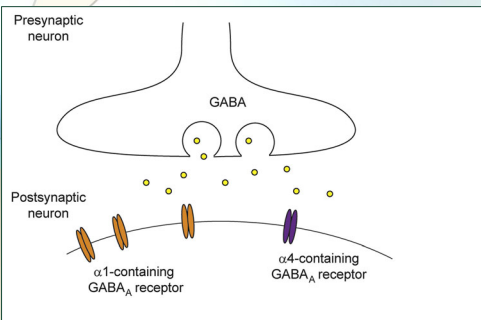
- EtOH inhibits iGlu receptors
 - NR1/2A or NR1/2B
 - and non-NMDA
- Decreased glu release

GABA

Acute drug actions of EtOH

EtOH increases Cl⁻ flux through GABA receptors

- Acute EtOH increases GABA_A receptor function
 - like benzodiazepines (BDZ), related to the anxiolytic, sedative and muscle relaxing properties of EtOH



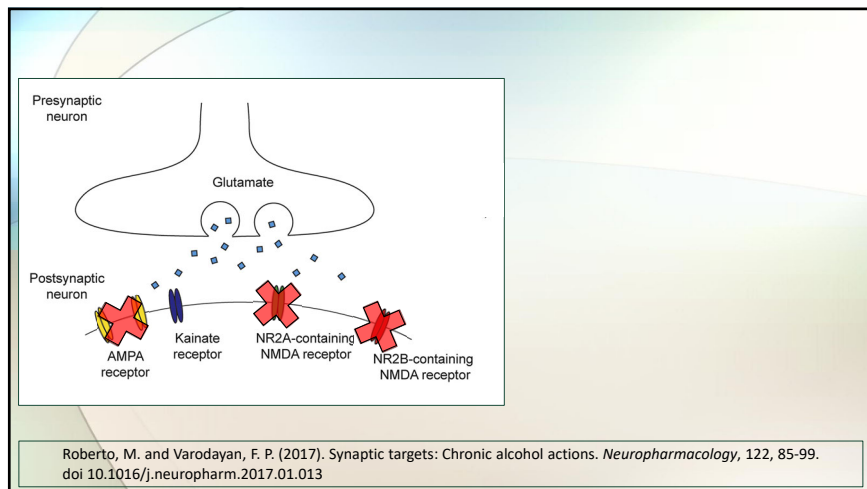
Roberto, M. and Varodayan, F. P. (2017). Synaptic targets: Chronic alcohol actions. *Neuropharmacology*, 122, 85-99. doi 10.1016/j.neuropharm.2017.01.013

Acute drug actions of EtOH

EtOH inhibits iGlu receptors & decreases Glu release

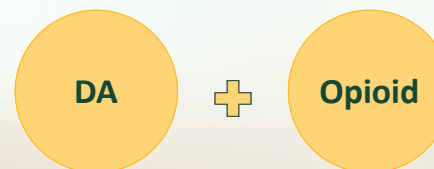
- Acute EtOH inhibits NMDA receptors & decreases LTP
 - “Blackouts” - disruption of memory consolidation

Chapter 10 – Alcohol



Acute EtOH modulates DA and opioid systems

Acute EtOH is positively reinforcing



Acute EtOH modulates DA and opioid systems

Acute EtOH is positively reinforcing

DA

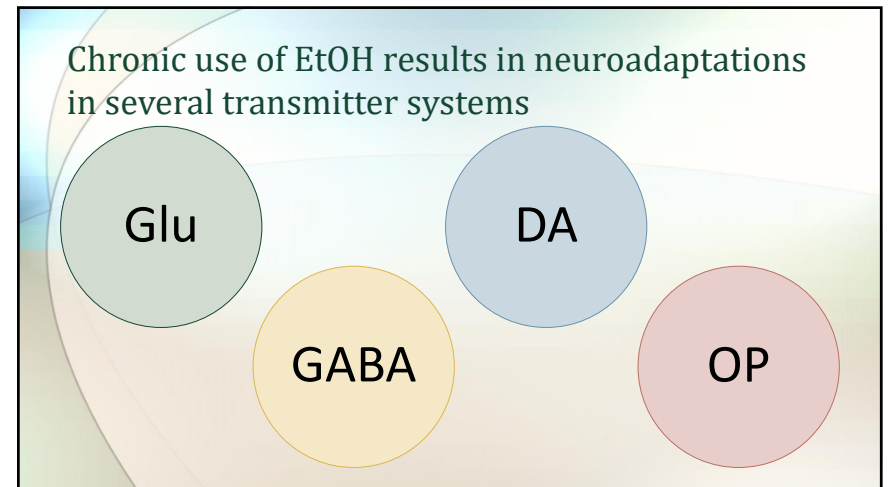
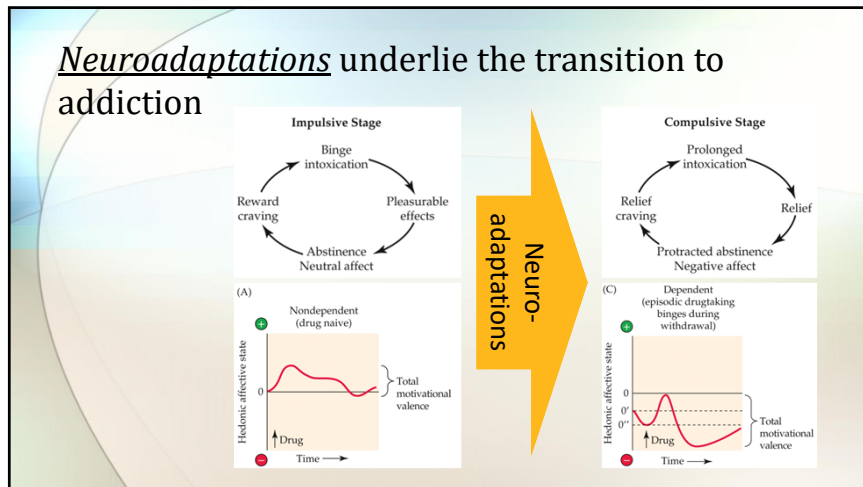
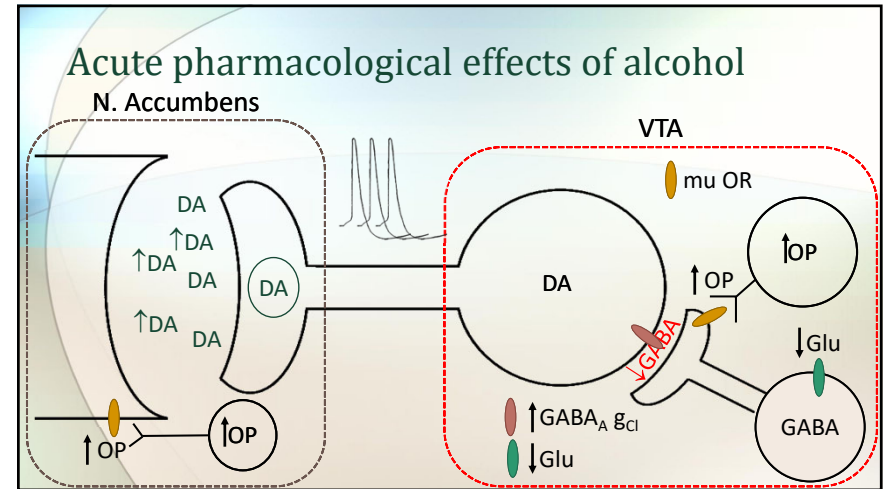
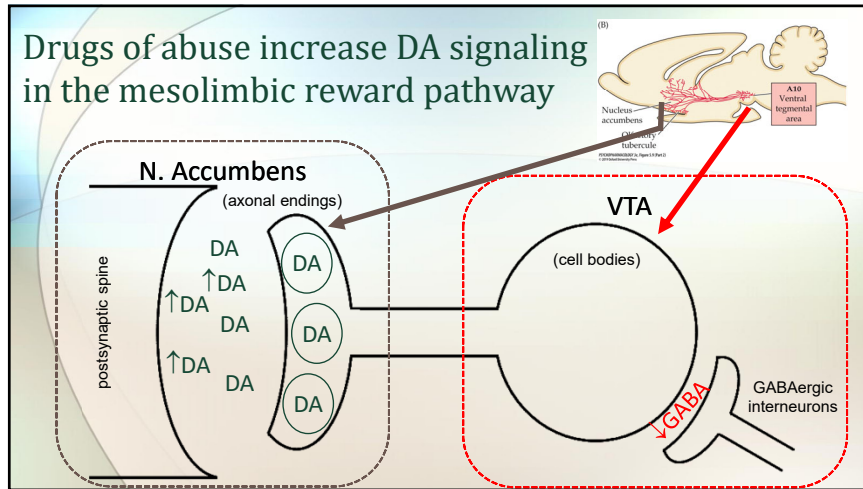
- EtOH increases firing rate of VTA neurons in the mesolimbic pathway, more DA released in N. Accumbens
- Rodents will self-administer EtOH directly into the VTA
- DA antagonists *reduce* self-administration of EtOH, but do not block it completely

Acute EtOH modulates DA and opioid systems

Acute EtOH is positively reinforcing

Opioid

- EtOH increases synthesis & release of enkephalins & endorphins
- promotes DA release in the mesolimbic pathway
- μ -OR antagonists (e.g., naltrexone) used to treat SUD's



Chronic use of EtOH results in neuroadaptations in glutamate transmission

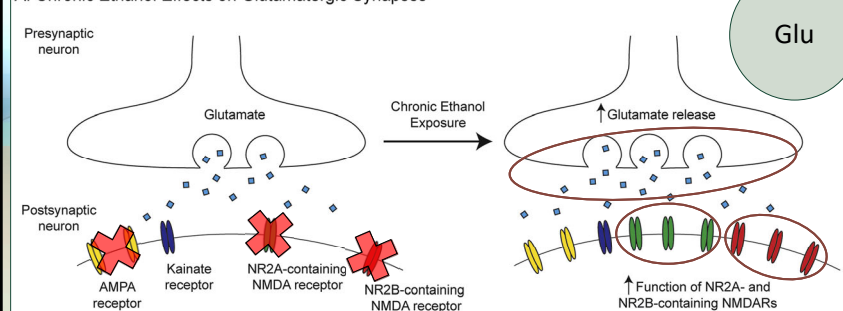
Glu

- *Acute* EtOH inhibits NMDA receptors & decreases LTP
- “Blackouts” - disruption of memory consolidation

- *Chronic* EtOH upregulates NMDA receptors and increases Glu release
- “Wet brain” - EtOH-induced excitotoxic damage
- following abstinence, increased seizure risk

Chronic use of EtOH results in neuroadaptations

A: Chronic Ethanol Effects on Glutamatergic Synapses



Roberto, M. and Varodayan, F. P. (2017). Synaptic targets: Chronic alcohol actions. *Neuropharmacology*, 122, 85-99. doi 10.1016/j.neuropharm.2017.01.013

Chronic use of EtOH results in neuroadaptations in GABA transmission

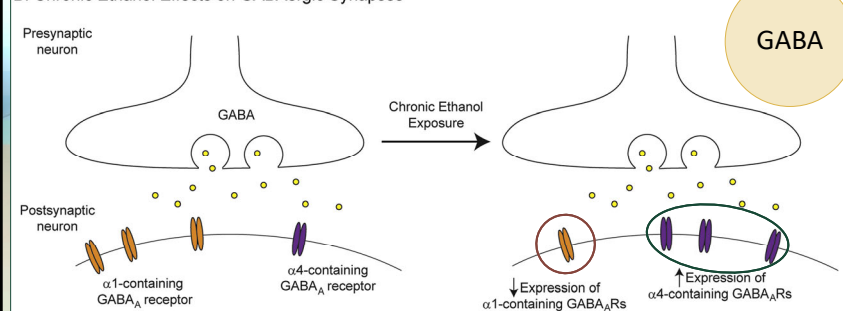
GABA

- *Acute* EtOH increases GABA_A receptor function
- like benzodiazepines (BDZ), related to the anxiolytic, sedative and muscle relaxing properties of EtOH

- *Chronic* EtOH associated with down-regulation of $\alpha 1$, but up-regulation of $\alpha 4$ and $\alpha 6$
- Increased sensitivity to convulsants; hyperexcitability; seizures; tremors
- BDZ's lose potency; contributes to cross-tolerance

Chronic use of EtOH results in neuroadaptations

B: Chronic Ethanol Effects on GABAergic Synapses



Roberto, M. and Varodayan, F. P. (2017). Synaptic targets: Chronic alcohol actions. *Neuropharmacology*, 122, 85-99. doi 10.1016/j.neuropharm.2017.01.013

Chronic use of EtOH results in neuroadaptations in dopamine transmission

DA

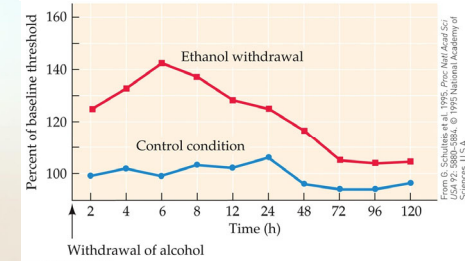
- ↓ AP firing in VTA neurons
- ↓ DA release in the N.Acc.
- tolerance; lower allostatic set-point

Chronic use of EtOH results in neuroadaptations in dopamine transmission

DA

- ↓ AP firing in VTA neurons
- ↓ DA release in the N.Acc.
- tolerance; lower allostatic set-point

Elevated ICSS threshold



Chronic use of EtOH results in neuroadaptations in opioid transmission

OP

- decreased opioid synthesis
- contributes to withdrawal-induced dysphoria

Acute pharmacological effects of alcohol

